



**Wednesday 10 May 2023
14.00 – 16.00 BST
Duration: 2 hours
Additional time: 30 minutes
Timed exam – fixed start time**

DEGREES of MSci, MEng, BEng, BSc, MA and MA (Social Sciences)

Information Retrieval (H)

COMPSCI4069

Answer all questions

This examination paper is an open book, online assessment and is worth a total of 60 marks.

1.

- (a) The following corpus of documents has been processed by an IR system where stemming is not applied:

Doc1: Real estate speculation is of interest.

Doc2: Interest rates are increasing interest in home costs.

Doc3: Students have no real interest in interest rates.

Doc4: As interest rates fall, the real estate market is heating up.

Doc5: The government is considering increasing interest rates.

- (i) Assume that the following terms are stopwords: an, as, and, are, do, in, is, of, not, the, up. Give the vector space model representations of documents Doc1 and Doc2, assuming that you are using (raw) term frequency to weight the terms. Show clearly the terms of your vectors as well as their weights. [2]
- (ii) Consider the following query Q:

Q= *interest rates*

Provide the vector space model representation of Q, showing both the terms as well as their weights. [1]

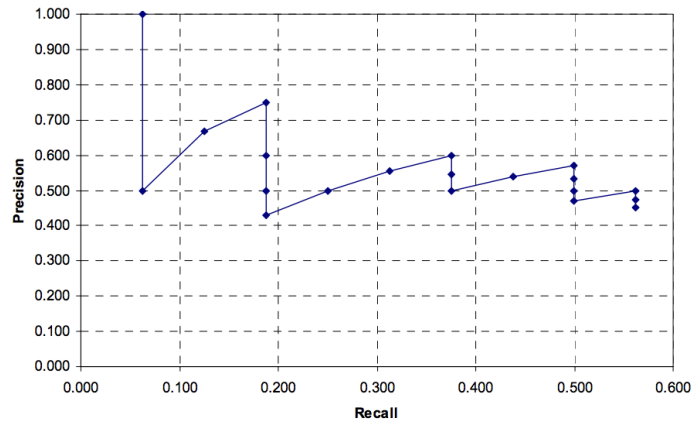
Compute the cosine similarity between the query Q and Doc1 as well as the cosine similarity between Q and Doc2. *Show your working.* [2]

- (iii) Assume the same list of stopwords as (i) above. Construct an inverted file for all the documents of the corpus, showing clearly the dictionary and posting list components. Your inverted file needs to store sufficient information for computing a simple $tf \cdot idf$ term weight, where $w_{ij} = tf_{ij} \cdot \log_2(N/df_i)$. [5]
- (iv) Assuming the use of a best match ranking algorithm, rank all documents of the corpus using their relevance scores for the following query:
real estate interest

Show your working. Note that $\log_2(1.5) = 0.3219$, $\log_2(1.6666) = 0.7369$, $\log_2(2.5) = 1.3219$ and $\log_2(5) = 2.3219$ (you may not need all of these).

[3]

- (b) Consider the following non-interpolated recall-precision graph, showing the performance of an IR system on a given query Q. For this query Q, the IR system has returned 20 documents. Assume that Q has 16 relevant documents in the ground truth, not all of which have been retrieved by the system for the query.



- (i) Compute the interpolated precision values for this query. Show your working.

[4]

- (ii) The IR system has returned 20 documents ranked from rank=1 to rank=20. In the tables below, indicate if the corresponding document at that rank was relevant (R) or non-relevant (X).

1	2	3	4	5	6	7	8	9	10

11	12	13	14	15	16	17	18	19	20

[2]

- (iii) Compute the Average Precision (AP) for query Q.

[1]

2.

- (a) Consider a collection A with 5000 documents with a lexicon size of 10000 words (number of unique words). Consider a collection B with 11250 documents and a lexicon size of 15000 words. Suppose that all documents have an average length of 200 words. We then add 2200 documents to each collection.

Use Heaps' Law with $\beta = 0.5$ to estimate how many additional unique terms we would add to A & B. Show your working.

Explain whether the obtained estimates are as you would expect, given the number of documents in A & B, and why?

[4]

- (b) Assume that a user's original query is **bargain books bargain DVDs really bargain books**. The user examines two documents, d_1 and d_2 . She judges d_1 , with the content **great books bargain bestsellers really bargain books** relevant and d_2 with content **big big bargain DVDs** non-relevant.

Assume that the IR system is using the vector space model with raw term frequency (i.e. with no IDF). Moreover, for relevance feedback, suppose the use of Rocchio's Relevance Feedback (RF) algorithm, with parameters $\alpha=1$, $\beta=0.75$, $\gamma=0.25$.

- (i) First, provide the original query vector. Moreover, provide the vector representations for d_1 and d_2 . You need to clearly show the terms of the vector in addition to the weights.

[2]

- (ii) Next, provide the reformulated query that will be run by the IR system after the application of Rocchio's RF algorithm. Terms in the reformulated query with negative weights can be dropped, i.e. their weights can be changed back to 0. *Show your working.*

[3]

- (iii) The users of the IR system requested a new feature in the user interface, where next to each returned document, they can click "Find documents like this one" to obtain more similar documents. You have been asked to re-use the Rocchio's algorithm to deploy such a feature and return similar documents. Explain which weight setting for α , β and γ you would use to deploy this new feature.

[2]

- (c) Consider the following six documents.

Doc1: I like Terrier Terrier Terrier

Doc2: I like Terrier and I like the course

Doc3: I like the course

Doc4: I don't like Terrier

Doc5: I like dogs

Doc6: I like puppies

Assume an IR system that uses the probabilistic Binary Independence Retrieval Model. Recall that in this model, the relevance score of a document to a query is computed using the following RSV formulae:

$$\sum_{i:d_i=1} \log \frac{p_i(1-s_i)}{s_i(1-p_i)}$$

Assume a large corpus (N=1000), and that the terms: like, Terrier, course, dogs and puppies only occur in the documents above (i.e. they do not occur in the rest of the 994 documents).

- (i) First, assume that there is no relevance information available to the system. Moreover, assume that p_i is a constant, namely $p_i=0.5$; i.e., half of all relevant documents will contain each query term. Now, consider the following query Q:

Q = Terrier

Calculate the Retrieval Status Value (RSV) of the six documents above for query Q, and rank the documents. Use log base 10. *Show your workings and justify your answer.*

[3]

- (ii) Next, assume that we have some information about term occurrences in the relevant and non-relevant documents for the above query Q. Indeed, through the use of relevance feedback, we were told that the only relevant documents for Q1 are: Doc1, Doc2, and Doc6. Everything else is non-relevant. Use this relevance information to compute the probabilities p_i and s_i . Then, calculate the RSV between the query Q and the six documents. Rank the documents for each query.

[4]

- (iii) Briefly provide the main take-away messages you might draw from your results in Parts (i) and (ii).

[2]

3.

- (a) For a particular query q , the multi-grade relevance judgements of all documents are $\{d1, 2\}, (d3,4), (d6,4), (d9,3), (d11,2), d31(1)\}$, where each tuple represents a document ID and a relevance judgement pair, and all the other documents are judged as non-relevant. Documents are judged on the scale 0-4 (0: not relevant – 4: highly relevant). An IR system returns the following ranked list with respect to this query (these are all results it returned for this query and in that order):

d31,d22,d3,d6,d15

Assume that $1/\log_2(\text{rank})$ is used as the discount factor. You might wish to note that: $\log_2(2) = 1$; $\log_2(3) = 1.59$; $\log_2(4) = 2$; $\log_2(5) = 2.32$; $\log_2(6) = 2.59$.

- (i) Compute the Normalised Discounted Cumulative Gain (NDCG) at each rank position for the given query. In your answer, you need to first provide the ideal DCG values for the perfect ranking for the given query, followed by the rest of your answer. Show your workings.

[4]

- (ii) DCG scores need to be normalised. Below is a quote from an AI-generated answer explaining why AP (and MAP) scores are not normalized. This answer is wrong. Provide a correct answer explaining why normalisation is not needed for AP/MAP scores?

“MAP (Mean Average Precision) is another measure used to evaluate the ranking performance of information retrieval systems. Unlike NDCG, MAP does not require normalization because it is a measure of average precision, which is a measure of the proportion of relevant items in the ranking list. MAP does not depend on the length of the list or the relevance scores of the items, so normalization is not needed.”

[2]

- (b) You oversee the architecture for a major search engine deploying learning-to-rank in a cascading architecture. All queries have a single relevant answer. Your current cascading architecture applies a learned model to re-rank a candidate set of 1000 documents. The result of the current architecture is that Recall@10 of 0.6 is achieved.

Stage	(ms)
Candidate Generation (1000 documents)	100
Feature Extraction for Existing Features (1000 documents)	100
Learned Model Reranking (1000 documents)	10
TOTAL	210ms

Your colleague has invented a new feature that can be combined into the learning-to-rank model – this feature increases precision - indeed, by applying this feature in conjunction with the existing features, it can ALWAYS place the relevant document within the first 10 ranks if the relevant document is in the candidate set. However, this new feature is very inefficient to apply, costing 200ms for 1000 documents.

Your boss demands that the search engine cannot become any slower overall. Discuss the impact on effectiveness of employing this new feature within the search engine architecture. What would your final recommendation be on the deployability of the new feature given the provided data?

You may refer to the following table detailing the Recall of the first stage candidate generation technique. You may assume that the efficiency of each ranking stage scales linearly the number of documents. *Justify your answer.*

	Recall @ cutoff									
Cutoff →	1000	900	800	700	600	500	400	300	200	100
Candidate Generation Method	0.70	0.68	0.66	0.60	0.55	0.50	0.40	0.30	0.20	0.10

[8]

- (c) Discuss why dense retrieval is advantageous for addressing search queries with vocabulary mismatch compared to a neural re-ranking setup. Your answer *must* include an example of a query and documents corresponding to a suitable information need scenario to illustrate your answer.

[6]